

WKB approximation

Paul Haskins

March 12, 2025

Wentzel-Kramer-Brillouin
construct approximate solutions to the time-independent Schrodinger equation.

we have the conservation of Energy (particle moving in 1D)

$$\frac{p^2}{2m} + V(x) = E \implies p^2(x) = 2m(E - V(x)) \quad (1)$$

Here $p(x)$ is the classical momentum not the qm operator.

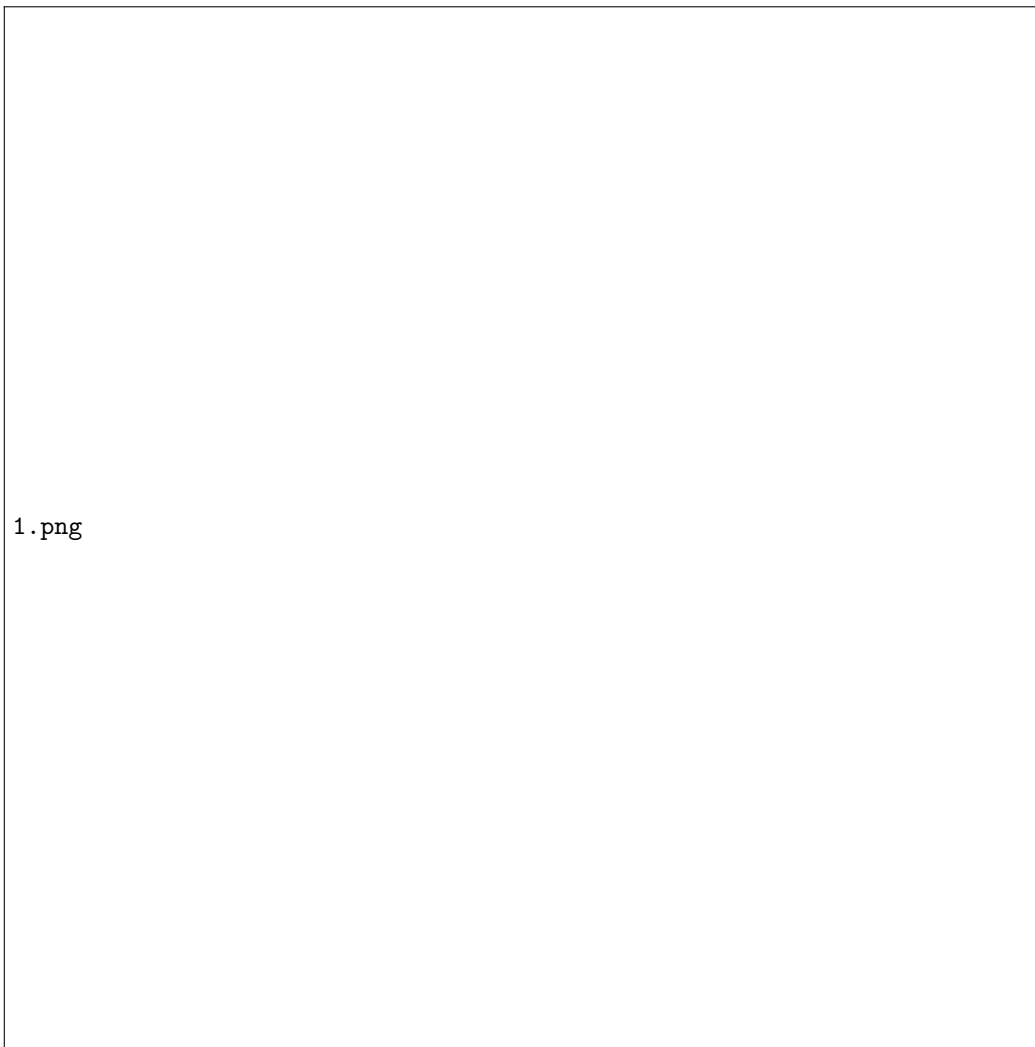
Quantum Mechanical Version

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) = E\psi \quad (2)$$

combining the two

$$\frac{d^2\psi}{dx^2} = -\frac{p^2(x)}{\hbar^2}\psi \quad (3)$$

Classically, particle always within the dashed box. In QM $\exists$ non-zero prob of finding particle outside the box.



1.png

Figure 1: Classical region 276 Griffiths

We have a general ansatz in classically allowed region.

$$\psi(x) = A(x) e^{i\phi(x)} \quad (4)$$

ψ is complex while A and ϕ are real.

$$\psi'(x) = A' e^{i\phi} + iA\phi' e^{i\phi} = (A' + iA\phi') e^{i\phi} \quad (5)$$

$$\phi'' = (A'' + iA'\phi' + iA\phi'') e^{i\phi} + i\phi (A' + iA\phi') e^{i\phi} = \left(A'' - A(\phi')^2 + i(2A'\phi' + A\phi'') \right) e^{i\phi} \quad (6)$$

Into the Schrodinger equation

$$A'' - A(\phi')^2 + i(2A'\phi' + A\phi'') = -\frac{p^2 A}{\hbar} \quad (7)$$

Taking the real part

$$A'' - A(\phi')^2 = -\frac{p^2 A}{\hbar} \quad (8)$$

The imaginary part: (assuming p is real i.e. in the classical region)

$$i(2A'\phi' + A\phi'') = 0 \quad (9)$$

Here we have a maths trick similar to integrating factor

$$\frac{d}{dx} (A^2 \phi') = A(2A'\phi' + A\phi'') \quad (10)$$

$$A^2 \phi' = \tilde{c}^2 \quad (11)$$

We have $\phi' \geq 0$ by convention

$$A = \frac{\tilde{c}}{\sqrt{\phi'}} \quad (12)$$

$$A'' + A \left(\frac{P^2}{\hbar^2} - (\phi')^2 \right) = 0 \quad (13)$$

We assume $\frac{A''}{A}$ is small

WKB approximation

$$\frac{P^2}{\hbar^2} - (\phi')^2 = 0 \quad (14)$$

$$\phi' = \pm \frac{P}{\hbar} \quad (15)$$

$$\phi = \pm \int p(x) dx \quad (16)$$

Thus we receive the WKB approximation in the classical region

$$\phi \approx \frac{c}{\sqrt{p(x)}} e^{\pm \frac{i}{\hbar} \int p(x) dx} \quad (17)$$